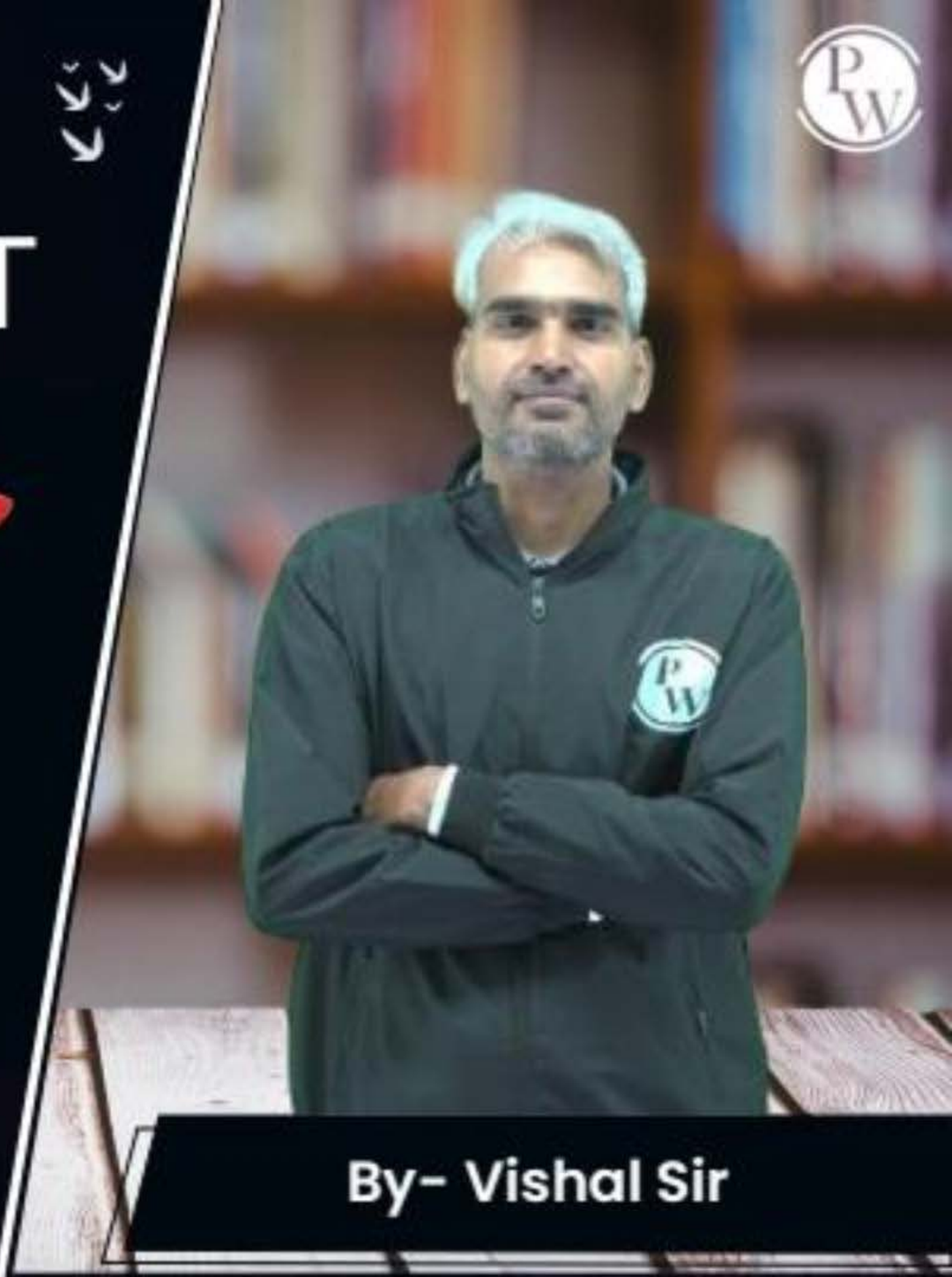
Computer Science & IT



Database Management System

Transactions and Concurrency control

Lecture No. 02



By- Vishal Sir

Recap of Previous Lecture



Topic — Transaction

Topic — ACID properties

Topic — Recovery management component

Slide

Topics to be Covered



Topic Schedule

Topic ACID properties

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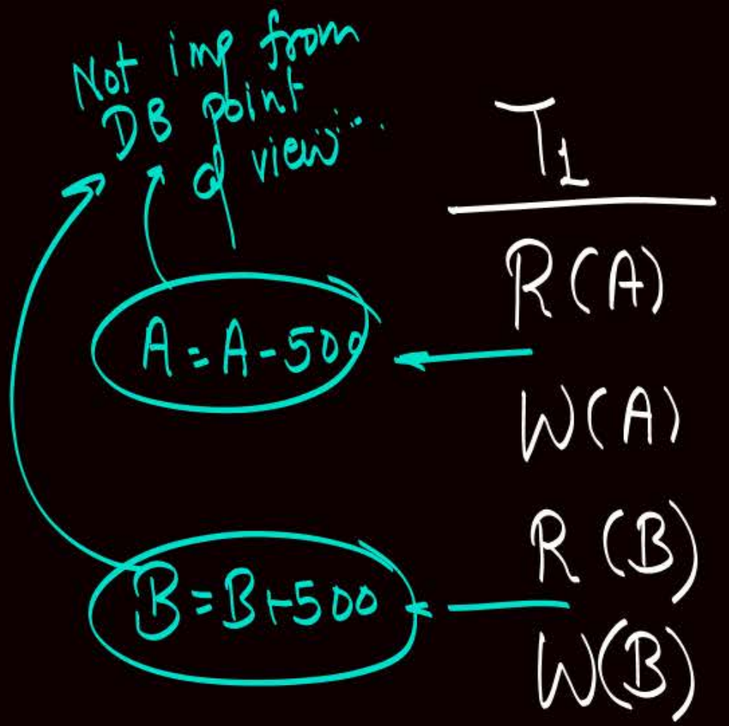


Topic : Isolation

Isolation says that if two or more transactions are executing Concurrently {simultaneously} then they must be unaware of each other

Consider two transactions : T_1 : Transfer 500 Rs from A to B

T_2 : Checking amount in A & B



T_1
R(A)
W(A)
R(B)
W(B)

T_2
R(A)
R(B)



Topic : Schedule

Time ordered sequence of operations of two or more transactions correspond to schedule

There are two types of schedule :-

1. Serial Schedule { one after another }
2. Concurrent Schedule { Interleaved execution of operations }



Topic : Serial Schedule

{ one after another }

Start the execution of a new transaction only after the Complete execution of previously started transaction

$R_i(A)$: Read of data item A by transaction T_i

$T_1 \rightarrow T_2$ { T_1 then T_2 }

T_1	T_2
$R_1(A)$	
$W_1(A)$	
$R_1(B)$	
$W_1(B)$	
	$R_2(A)$
	$R_2(B)$

increasing time

$T_2 \rightarrow T_1$ { T_2 then T_1 }

T_1	T_2
	$R_2(A)$
	$R_2(B)$
$R_1(A)$	
$W_1(A)$	
$R_1(B)$	
$W_1(B)$	

increasing time



Topic : Serial Schedule

* Serial schedule will always satisfy isolation.
  {Because transactions are not executing Concurrently}
* With 'n' transactions, n! different serial schedules are possible

$n * (n-1) * (n-2) | \cdots | 3 * 2 * 1$

* Serial schedule will always satisfy isolation condition, but throughput will be very low with serial schedules.



Topic : Concurrent Schedule

* Interleaved execution of operations of two or more transaction corresponds to Concurrent schedule.

Eg:

S_1

T_1	T_2
$R_1(A)$	
$W_1(A)$	
	$R_2(A)$
$R_1(B)$	
$W_1(B)$	
	$R_2(B)$

S_2

T_1	T_2
$R_1(A)$	
$W_1(A)$	
	$R_2(A)$
	$R_2(B)$
$R_1(B)$	
$W_1(B)$	



Topic : Concurrent Schedule

- Throughput of the system will increase with Concurrent execution of transactions, but Concurrent execution may dissatisfy isolation Condition
- For isolation to be satisfied behaviour of Concurrent schedule must be equivalent to at least one of the Serial schedule

  → For behaviours of two schedules to be equivalent
  ① Every read opn must be same in both schedules
  & ② Final update (write) of every data item must also be same in both schedules



Topic : Serializable schedule

* Concurrent schedule whose behaviour is equivalent to atleast one of the serial schedule is called Serializable schedule



Topic : Isolation

For isolation to be satisfied our schedule must be serializable schedule.

i.e. its behaviour must be equivalent to atleast one of the serial schedule

eg. (S1) ✓

T_1	T_2
1000 ← $R_1(A)$	
500 ← $W_1(A)$	
	$R_2(A)$ → 500
0 ← $R_1(B)$	
500 ← $W_1(B)$	
	$R_2(B)$ → 500

- Every Read is same as serial schedule $T_1 \rightarrow T_2$
- Both A & B are finally updated by transaction T_1 in serial schedule $T_1 \rightarrow T_2$ and in schedule S1 also A & B are finally updated by T_1

Let initially A = 1000, B = 0

Serial schedule $T_1 \rightarrow T_2$

T_1	T_2
1000 ← $R_1(A)$	
500 ← $W_1(A)$	
0 ← $R_1(B)$	
500 ← $W_1(B)$	
	$R_2(A)$ → 500
	$R_2(B)$ → 500

- Every Read in S_1 is same as $T_1 \rightarrow T_2$, (and)
- Final update of every data item in S_1 is also same as $T_1 \rightarrow T_2$

∴ S_1 is equivalent to $T_1 \rightarrow T_2$

∴ S_1 is a serializable schedule

∴ w.r.t. S_1 isolation is satisfied.

eg. (S1)

Let initially A = 1000
B = 0

T_1	T_2
1000 ← $R_1(A)$	
500 ← $W_1(A)$	
	$R_2(A)$ → 500
0 ← $R_1(B)$	
500 ← $W_1(B)$	
	$R_2(B)$ → 500

This two reads are not same
∴ S_1 is not equivalent to serial schedule $T_2 \rightarrow T_1$

Serial schedule $T_2 \rightarrow T_1$

T_1	T_2
	$R_2(A)$ → 1000
	$R_2(B)$
$R_1(A)$	
$W_1(A)$	
$R_1(B)$	
$W_1(B)$	

We observed,

$S_1 \equiv T_1 \rightarrow T_2$

&

$S_1 \not\equiv T_2 \rightarrow T_1$

S_1 is equivalent to at least one serial schedule with transactions T_1 & T_2

$\therefore S_1$ is a serializable schedule

Hence S_1 satisfies isolation

eg. S_2

T_1	T_2
1000 ← $R_1(A)$	
500 ← $W_1(A)$	
	$R_2(A)$ → 500
	$R_2(B)$ → 0
$R_1(B)$	
$W_1(B)$	

Let initially A = 1000, B = 0

Serial schedule $T_1 \rightarrow T_2$

T_1	T_2
1000 ← $R_1(A)$	
500 ← $W_1(A)$	
0 ← $R_1(B)$	
500 ← $W_1(B)$	
	$R_2(A)$ → 500
	$R_2(B)$ → 500

$R_2(B)$ are not same in schedule S_2 & serial schedule $T_1 \rightarrow T_2$

$\therefore S_2 \not\equiv T_1 \rightarrow T_2$

eg. (S2)

Let initially A = 1000
B = 0

T_1	T_2
1000 ← $R_1(A)$	
500 ← $W_1(A)$	
	$R_2(A)$ → 500
	$R_2(B)$
0 ← $R_1(B)$	
500 ← $W_1(B)$	

This time reads are not same

∴ S_2 is not equivalent to serial schedule $T_2 \rightarrow T_1$

$S_2 \neq T_2 \rightarrow T_1$

Serial schedule $T_2 \rightarrow T_1$

T_1	T_2
	$R_2(A)$ → 1000
	$R_2(B)$
$R_1(A)$	
$W_1(A)$	
$R_1(B)$	
$W_1(B)$	

We observed,

$S_2 \not\equiv T_1 \rightarrow T_2$

&

$S_2 \not\equiv T_2 \rightarrow T_1$

S_2 is not equivalent to any of the serial schedule with transactions T_1 & T_2

i.e. S_2 is not a serializable schedule

Hence S_2 does not ensure isolation

Note:-

① Concurrency Control Component is responsible for avoiding the execution of non-serializable schedule.

∴ Concurrency Control Component is responsible for ensuring isolation

Note: Every serial schedule is serializable schedule, but every serializable schedule need not be a serial schedule



Topic : Consistency

Consistency says that during and after the execution of transactions the database should remain Consistent

For Consistency {
① Schedule must be recoverable schedule { more precisely strict recoverable }
② Schedule must be Serializable Schedule
}

* Both Concurrency Control Component and recovery management Components are responsible for ensuring Consistency



2 mins Summary

Topic Serial schedule & concurrent schedule

Topic Isolation { Serializable Schedule }

Topic Consistency

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THANK - YOU

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